
DS5104: STATISTICAL METHODS FOR DATA SCIENCE

DEADLINE: 29/07/2026

FINAL ASSESSMENT

INDEX NUMBER: _____

SECTION A: GENERAL INSTRUCTIONS & GUIDELINES

- Answer all questions in both Part A and Part B.
- Use appropriate statistical software (e.g., Python or R) where computation is required.
- Clearly state all null (H_0) and alternative (H_1) hypotheses prior to conducting statistical tests.
- All hypothesis tests must be conducted at a significance level of $\alpha = 0.05$ unless explicitly stated otherwise.
- Provide formal statistical interpretations alongside numerical outputs (test statistics, degrees of freedom, and p -values).

Part A: FitPulse Health Dataset Analysis

1. (15 points) Normality & Descriptive Metrics

Using the variables `daily_steps_avg` and `age` from `fitpulse_health_dataset.csv`, evaluate distributional characteristics.

- (4 points) Formulate H_0 and H_1 for testing the normality of `daily_steps_avg` via the Shapiro-Wilk test.
- (11 points) Perform the Shapiro-Wilk test at $\alpha = 0.05$:
 1. Report the test statistic (W) and corresponding p -value.
 2. State your statistical decision regarding H_0 .
 3. Determine whether to summarize the variable using **Mean (SD)** or **Median (IQR)**.
 4. Justify your metric choice based on distribution characteristics.

2. (10 points) Count Data Fundamentals

The outcome `supplement_purchases` measures non-negative annual user orders.

- (4 points) Explain two primary theoretical reasons why OLS linear regression fails for non-negative count data.
- (6 points) Identify a suitable count distribution, state its core assumptions, and explain how overdispersion invalidates it.

3. (15 points) A/B Experimentation

An A/B test (`ab_test_group`: *Control* vs. *Variant_B*) measures impact on `daily_steps_avg`.

- (a) (3 points) State H_0 and H_1 for an Independent Two-Sample t -test comparing mean step counts between groups.
- (b) (8 points) Execute the test, reporting test statistic (t), df , p -value, statistical decision, and a deployment recommendation.
- (c) (4 points) Contextualize Type I (α) and Type II (β) errors in terms of business impact for this product launch.

4. (15 points) **Categorical Association**

Evaluate customer churn across subscription tiers (`subscription_tier`: *Free*, *Basic*, *Premium*).

- (a) (15 points) Conduct a Chi-Square (χ^2) Test of Independence against `churned_within_6mo`. State hypotheses, present the expected frequencies table, and report df , χ^2 statistic, p -value, and retention insights.

5. (20 points) **Multiple Linear Regression**

Consider the multiple regression model predicting daily activity:

$$\widehat{\text{steps}} = \hat{\beta}_0 + \hat{\beta}_1(\text{age}) + \hat{\beta}_2(\text{sleep_quality})$$

- (a) (6 points) Fit the model and interpret coefficients $\hat{\beta}_0$, $\hat{\beta}_1$, and $\hat{\beta}_2$ in context.
- (b) (4 points) Assess predictor significance using partial t -tests and p -values.
- (c) (10 points) Define multicollinearity, explain how Variance Inflation Factor (VIF) detects it, and discuss its impact on standard errors.

6. (25 points) **Logistic Regression & Classification**

Model churn probability (`churned_within_6mo`: 1 = Yes, 0 = No):

$$\ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1(\text{age}) + \beta_2(\text{Premium_Tier})$$

- (a) (5 points) Explain why the logit link function is preferred over linear models for binary outcomes.
- (b) (10 points) Compute and interpret Odds Ratios ($\text{OR} = e^{\beta}$) for both predictors. Clarify if Premium tier reduces churn odds.
- (c) (10 points) Evaluate overall model performance using Accuracy, Precision, Recall, and ROC-AUC metrics.

Part B: Seaborn `tips` Dataset Analysis ($N = 244$)

1. (15 points) **Distributional Assessment**

Analyze total dining expenditure (`total_bill`).

- (a) (5 points) Formulate hypotheses and perform a Shapiro-Wilk test on `total_bill`.
- (b) (10 points) Report statistic (W), p -value, and decision. Justify choosing **Mean (SD)** vs. **Median (IQR)**.

2. (10 points) **Party Size Modeling**

Analyze customer party size (`size`).

- (a) (5 points) Explain why OLS regression is unsuitable for discrete party counts like `size`.

- (b) (5 points) Describe the Poisson assumption of equidispersion and state alternative models when variance exceeds the mean.

3. (15 points) **Gender Spend Comparison**

Compare spending between male and female bill payers (`sex` vs. `total_bill`).

- (a) (10 points) State hypotheses and conduct an Independent Two-Sample t -test (t , df , p -value, decision).
- (b) (5 points) Explain the practical consequences of committing a Type I vs. Type II error for restaurant management.

4. (15 points) **Analysis of Variance & Independence**

- (a) (8 points) Conduct a One-Way ANOVA testing mean `total_bill` across operating days (`Thur`, `Fri`, `Sat`, `Sun`). Provide hypotheses, ANOVA table summary (F , df , p -value), conclusion, and recommended post-hoc test.
- (b) (7 points) Conduct a Chi-Square (χ^2) Test of Independence between `sex` and `smoker` (df , χ^2 , p -value, conclusion).

5. (20 points) **Tip Prediction Regression**

Consider the tip prediction model:

$$\widehat{\text{tip}} = \beta_0 + \beta_1(\text{total_bill}) + \beta_2(\text{size})$$

- (a) (8 points) Fit the model and interpret slope parameters $\hat{\beta}_1$ and $\hat{\beta}_2$.
- (b) (5 points) Evaluate individual predictor significance using partial t -tests.
- (c) (7 points) Compute VIF values for predictors and discuss collinearity risk between bill size and table count.

6. (25 points) **High-Tip Binary Logistic Model**

Define binary outcome `high_tip` = 1 if `tip` $\geq$ \$3.00, else 0. Fit logit model:

$$\ln \left(\frac{p}{1-p} \right) = \beta_0 + \beta_1(\text{total_bill}) + \beta_2(\text{smoker})$$

- (a) (5 points) Justify logistic regression over linear probability models for `high_tip`.
- (b) (10 points) Report model estimates: coefficients ($\hat{\beta}$), standard errors (SE), z -scores, and p -values.
- (c) (10 points) Derive and interpret Odds Ratios ($e^{\hat{\beta}}$) for `total_bill` and `smoker` with actionable staff insights.